Dependency-Aware Multi-Phase Liver Tumor Segmentation with Explicit Phase Interaction Modeling

Areeba Naveed (27100239) Haider Wajahat (27100252)

GitHub Repository Link: `https://github.com/areebanaveed-12/Phase-Dependency-Liver-Segmentation`

Abstract

Liver tumor segmentation from CT is clinically important for diagnosis, treatment planning, and longitudinal monitoring. Our project began with a standard single-phase U-Net baseline on LiTS and then progressed toward a multi-phase CT setting motivated by our SOA survey, which identified a gap between implicit fusion methods and explicit modeling of phase relationships. We first adapted the pipeline to Harvard multi-phase CT data with a four-phase attention U-Net baseline. We then proposed a dependency-aware segmentation model that preserves phase-specific representations, learns per-phase importance gates, and adds an explicit pairwise cross-phase interaction term before decoding. The final system uses 2.5D input with three neighboring slices per phase, tumor-aware sampling, a combined focal CE + Dice + focal Tversky objective, and validation-tuned inference thresholds. On the final 100-patient Harvard experiment, the model achieved a liver Dice score of 0.9435 and a tumor Dice score of 0.6975. A major contribution of the study is a complete combinational dependency analysis on the fixed trained model across all 15 possible non-empty phase combinations. This analysis showed that the best two-phase subset, PVP+DP, achieved a tumor Dice of 0.7209 and the best three-phase subset, AP+PVP+DP, achieved 0.7109, both outperforming the all-phase setting at 0.6975.

1. Introduction

Liver tumor segmentation from CT lies at the intersection of medical image analysis and clinical decision support. Accurate delineation of the liver and intrahepatic tumors matters because it affects diagnosis, treatment planning, volumetric assessment, and longitudinal monitoring. In routine abdominal imaging, the problem is not limited to a single scan. Multi-phase CT acquires the liver across non-contrast (NC), arterial phase (AP), portal venous phase (PVP), and delayed phase (DP), and each phase reveals different enhancement behavior and lesion conspicuity (Wu et al., 2024; Gul et al., 2022; Chernyak et al., 2018).

This is where our SOA survey became important. The survey helped establish the problem domain, summarize the major segmentation baselines, and identify the gap that still remained. Classical encoder–decoder baselines such as U-Net are effective for dense medical segmentation (Ronneberger et al., 2015), and more recent multi-phase methods improve results through attention, fusion, adaptive weighting, or missing-phase compensation (Wu et al., 2024; Gul et al., 2022; Zhang et al., 2021b; Xu et al., 2021; Liu et al., 2024; Hu et al., 2024). However, the survey showed that many methods still treat phase relationships implicitly. They use multiple phases, but they do not directly study which combinations are most useful or how phases interact with one another.

That gap motivated our main project idea. Instead of asking only whether multi-phase input improves segmentation, we asked whether a model could preserve phase-specific information long enough to reason about cross-phase dependence explicitly. We therefore designed the project as a progression: Phase 3 established a corrected single-phase LiTS baseline, Phase 4 adapted the pipeline to Harvard multi-phase CT, and Phase 5 introduced a dependency-aware model together with a complete combinational dependency analysis on the fixed trained model.

Goals of this paper:

1. Establish a reliable segmentation baseline using a corrected single-phase U-Net on LiTS so that later improvements can be interpreted against a concrete starting point.

2. Adapt the pipeline from single-phase CT to a realistic four-phase Harvard setting with phase-aware preprocessing, caching, and patient-wise evaluation.

3. Implement a dependency-aware multi-phase architecture that preserves per-phase representations, learns

phase-specific importance, and models explicit cross-phase interactions.

4. Evaluate not only final Dice scores but also phase behavior itself through a complete study of all 15 possible non-empty phase combinations.

5. Show that certain reduced phase subsets can outperform the all-phase configuration, thereby demonstrating why explicit dependency analysis is a meaningful contribution rather than a cosmetic add-on.

2. Related Work

U-Net remains the canonical encoder–decoder baseline for dense medical image segmentation (Ronneberger et al., 2015), and LiTS remains the most widely recognized public benchmark for liver and liver tumor segmentation (Bilic et al., 2023). Task-specific liver segmentation work such as H-DenseUNet (Li et al., 2018), together with modern segmentation systems such as nnU-Net (Isensee et al., 2021), UNETR (Hatamizadeh et al., 2022), and Swin-UNETR (Tang et al., 2022), established the importance of strong feature hierarchies, context modeling, and robust training pipelines.

Surveyed multiphase methods have shown that richer CT input can improve segmentation through channel fusion, phase weighting, spatial aggregation, or image synthesis (Wu et al., 2024; Gul et al., 2022; Zhang et al., 2021b; Xu et al., 2021; Raju et al., 2020; Liu et al., 2024). However, our SOA survey concluded that the major open issue is no longer whether multiple phases help, but how their relationships should be modeled. Many systems can reweight phases, yet they do not make phase dependencies themselves the main object of learning and interpretation.

The related literature also revealed a practical issue: the strongest truly multi-phase liver lesion datasets used in recent papers are often private, while the public LiTS benchmark is not a dedicated four-phase dataset (Bilic et al., 2023; Liu et al., 2024). This creates a reproducibility gap between the algorithms that are conceptually closest to our problem and the data that are easiest to benchmark publicly.

Methods reviewed in our survey span a broad range of strategies. Some focus on feature aggregation with attention or adaptive fusion; others rely on synthesis to complete missing phases; still others encourage cross-phase interaction at the spatial level (Xu et al., 2021; Liu et al., 2024; Wu et al., 2023). Missing-view and incomplete-modality work further shows that robustness can be improved through heterogeneity-aware adaptation, multimodal transformers, or synthesis-based completion (Raju et al., 2020; Zhang et al., 2022; Chen et al., 2024). The common pattern is that multiphase evidence is useful, but the mechanism by which a model should represent phase agreement, contradiction, or complementarity remains under-specified (Zhang et al., 2021b).

For our report, that distinction matters because our final contribution is not simply another attention U-Net. The novelty target identified in the survey was to preserve the identity of each phase long enough that the network can explicitly reason about interactions among them. This is the standard against which the final notebook should be judged.

That observation directly motivated our final design. We wanted a model that preserved phase-specific features, reasoned over how phases reinforce one another, and exposed interpretable evidence about phase importance and phase subset behavior.

3. Methodology

This methodology section is organized as a stacked progression of improvements. Each phase reused the working parts of the previous pipeline and then introduced a more realistic dataset setting, a stronger representation, or a more explicit modeling assumption. In that sense, the methodology follows a pyramid: Phase 3 established the baseline segmentation machinery, Phase 4 converted that machinery into a real multi-phase pipeline, and Phase 5 introduced the final dependency-aware architecture together with the main dependency experiments.

Phase 3 baseline on LiTS. The corrected baseline notebook implemented a conventional 2D U-Net with four encoder stages, a bottleneck, and a symmetric decoder. The input channel count was configurable, but the baseline used one HU-windowed CT slice per example. The fixed LiTS pipeline used a volume-level split with 35 training volumes, 8 validation volumes, and 8 test volumes, producing 4,978 training slices, 833 validation slices, and 930 test slices after liver-slice filtering. Training used a combined cross-entropy and Dice loss, and the final corrected test result reached 0.9111 liver DSC and 0.3360 tumor DSC.

Phase 4 first improvement on Harvard multi-phase CT. The next notebook kept the basic segmentation training structure from Phase 3 but changed the problem setting completely. Instead of single-phase LiTS slices, the pipeline now had to ingest four distinct CT phases per patient. Each cached sample stored a 4-channel stack corresponding to NC, AP, PVP, and DP, together with a three-class segmentation mask, a presence vector, and patient metadata. This stage introduced TAR extraction, phase identification, cache manifests, patient-wise splitting, and tumor-aware oversampling. The active model in this notebook was a four-phase *BaselineAttentionUNet* (Oktay et al., 2018).

Phase 5 final dependency-aware model. The final notebook retained the Harvard multi-phase framework from Phase 4 but upgraded both the data representation and the modeling strategy. The dataset expanded from 50 to 100 patient archives, and the input representation moved from one slice per phase to a 2.5D formulation with three neighboring slices per phase, resulting in 12 total input channels. This preserved local axial context without moving to a full 3D pipeline and made phase reasoning more spatially informed.

The Harvard pipeline also added several practical steps that were not present in the single-phase baseline. The notebook discovers data automatically from cloud storage, stages archives locally for stable access, validates cache metadata to avoid stale preprocessed slices, and stores every training example as a compressed NPZ file containing the image tensor, mask, phase presence vector, patient identifier, and slice index. These engineering steps are part of the methodology because the final model depends on stable multiphase alignment and repeatable preprocessing.

Slice construction became an important methodological decision. Rather than sampling all slices uniformly, the final notebook keeps tumor-bearing slices whenever possible and then adds nearby liver-context slices around them. This strategy improves the effective lesion-to-background ratio during training while still preserving enough organ anatomy for the decoder to learn coherent liver boundaries.

Architecturally, the progression from Phase 4 to Phase 5 is the main theoretical move of the paper. The Phase 4 *BaselineAttentionUNet* still treated the four phases as a single jointly stacked input stream. Its decoder used attention-gated skip connections, but the feature extractor itself did not preserve phase identity beyond the raw input channels. The final *DependencyAttentionUNet* changes this by introducing a shared per-phase encoder. Each phase-specific input group is processed independently to produce a deep feature map and skip tensors before any fusion happens.

The shared per-phase encoder is important for two reasons. First, it preserves a distinct representation for each phase, which is necessary if phase relationships are to be modeled explicitly rather than absorbed into a generic feature tensor. Second, it keeps the architecture parameter-efficient because the same encoder weights are reused across NC, AP, PVP, and DP. This gives the model a principled way to compare phases at the feature level while still learning a common representation space (Zhang et al., 2022).

After encoding, the model computes a scalar gate for each phase from the corresponding bottleneck feature. These gates act as learned phase-importance weights and are used to form a normalized weighted average of both the bottleneck features and the skip tensors. The implementation also clamps the effective gate range to $[0.15, 1.0]$ so that no phase collapses completely during training. This means the decoder does not receive a blind concatenation of phase features; it receives a fused representation whose composition depends on what the model has learned about phase usefulness for that sample.

The main novelty is an explicit pairwise interaction term. For every pair of phases i and j, the model computes an elementwise interaction feature $f_i \odot f_j$. These pairwise interaction maps are averaged, projected back into the fused bottleneck space, and added to the gated fused representation before decoding. The practical interpretation is that the model is encouraged to learn where phases agree, reinforce one another, or jointly highlight structures that would be weaker in a single stream. This is the core architectural step that moves the project from implicit multi-phase fusion to explicit interaction modeling (Zhang et al., 2021b; Xu et al., 2021; Liu et al., 2024).

The decoder then operates on this interaction-aware fused representation using attention-gated skip information (Oktay et al., 2018). This preserves one advantage of the Phase 4 baseline, namely stronger skip refinement, while changing the information passed into the decoder. So the final architecture is not a complete replacement of earlier design ideas; it is an extension that preserves useful components and inserts explicit dependency modeling where the earlier baseline was weakest.

Loss and inference. The final training objective was a tumor-focused combination of focal cross-entropy (Lin et al., 2017), weighted Dice loss, and tumor focal Tversky loss (Abraham & Khan, 2019), mixed at weights 0.25/0.35/0.40. The checkpoint selection criterion was tumor-weighted ($0.35 \cdot \text{liver} + 0.65 \cdot \text{tumor}$). Inference used horizontal-flip test-time augmentation (TTA), validation-tuned liver and tumor thresholds, and post-processing to remove very small tumor components and fill organ holes. Light phase dropout during training regularized the phase-fusion process and made the learned representation more suitable for the all-15-subset analysis carried out later on the fixed trained model.

3.1. Phase 3: Baseline U-Net on LiTS

Phase 3 established the minimum viable segmentation pipeline for the project. The corrected baseline was a standard U-Net operating on one CT slice at a time after applying a liver-focused HU window. This phase was important because it let us validate our choice of labels, metrics, training loop, checkpointing logic, and qualitative inspection workflow before moving to the harder multi-phase setting.

Unlike the earlier mistaken submission notebook, the fixed

Phase 3 baseline used a volume-level split, which is much safer than a slice-level split. The corrected notebook used 35 training volumes, 8 validation volumes, and 8 test volumes, yielding 4,978 train slices, 833 validation slices, and 930 test slices after filtering for liver-containing images. That makes the fixed baseline a more reliable reference for project progression, even though it remains on LiTS rather than the later Harvard dataset.

Architecturally, the Phase 3 U-Net used a plain encoder–bottleneck–decoder pathway without phase reasoning. Its corrected test result was 0.9111 liver DSC and 0.3360 tumor DSC. Its importance in the project is therefore conceptual and procedural: it tells us what a straightforward segmentation model can achieve when no explicit phase decomposition or dependency mechanism is available (Li et al., 2018; Isensee et al., 2021).

3.2. Phase 4: Harvard Multi-Phase Baseline

Phase 4 marked the first true alignment between implementation and project idea. We no longer worked with a public single-phase proxy dataset; instead, we moved to Harvard archives that contained all four CT phases and phase-specific liver and tumor masks. This phase required solving not only model design but also data engineering issues that were absent from the LiTS baseline.

The preprocessing code for Phase 4 had to identify CT phases from filenames, separate liver and tumor masks, validate volume geometry, and build a cache that could be safely reused between runs. Each resulting NPZ sample stored a 4-channel multi-phase image, the merged 3-class mask, the phase-presence vector, and patient metadata. This was the first stage where the project became genuinely phase-aware at the data level.

The active model in Phase 4 was a *BaselineAttentionUNet*. It was already stronger than the plain LiTS U-Net in terms of skip-connection refinement because it used attention gates in the decoder (Oktay et al., 2018). However, it still treated the four phases as a jointly stacked input stream. In other words, Phase 4 was a multi-phase baseline, but not yet a dependency-aware one (Xu et al., 2021; Liu et al., 2024).

This phase also established patient-wise train, validation, and test splits, which is methodologically important in medical imaging. Once the same patient can appear in multiple slices, patient-level separation becomes much safer than random slice-level splitting. The Harvard baseline therefore represented an implementation improvement in both data realism and evaluation rigor.

3.3. Phase 5: Final Dependency-Aware Submission

Phase 5 is the final project submission and the notebook most closely aligned with the SOA-survey novelty. The major conceptual change is that phases are no longer fused immediately. Instead, the network preserves phase-specific representations until after each phase has been encoded, gated, and incorporated into an explicit interaction term.

Table 1. Harvard data scale and split progression from the baseline to the final submission.

Property	Phase 4	Phase 5
Patient archives	50	100
Train / val / test pts	34 / 8 / 8	70 / 15 / 15
Train / val / test slices	721 / 153 / 168	2276 / 462 / 474
Input channels	4	12
Phase context	1 slice / phase	2.5D, 3 / phase

The 2.5D input formulation is especially important here. Each phase contributes three neighboring slices, so the model can see local axial continuity within each phase before reasoning across phases. This design gives the model more context than a single 2D slice while avoiding the computational burden of full 3D segmentation.

The final notebook also tightened the optimization procedure. Tumor slices were oversampled more strongly, the loss was shifted toward tumor-sensitive terms, early stopping was added, and the final test predictions were decoded with validation-tuned thresholds rather than plain argmax alone. Concretely, the optimization used a 0.25/0.35/0.40 mixture of focal cross-entropy, weighted Dice, and tumor focal Tversky losses, together with a tumor oversampling weight of 12.0. These choices reflect the practical reality that tumor segmentation is much harder than liver segmentation and requires more targeted optimization.

Finally, Phase 5 is where the project became interpretable in a way that matters for the survey claim. The network reports average phase-gate activations, and the notebook performs a complete combinational dependency analysis on the fixed trained model by evaluating all 15 possible non-empty phase combinations. This means the project is not only presenting final Dice values but also presenting direct evidence about phase dependence itself, including higher-order three-phase behavior.

4. Results

4.1. Phase-by-Phase Result Progression

The clearest project trend is the transition from the four-phase Harvard baseline to the final dependency-aware system. The first Harvard baseline achieved 0.8896 liver DSC and 0.5287 tumor DSC on the 50-case setup. The final system achieved 0.9435 liver DSC and 0.6975 tumor DSC on the 100-case setup. Because the final pipeline also introduced 2.5D context, stronger slice selection, richer losses, tuned thresholds, and post-processing, the gain should be in-

Table 2. Final experimental setup and inference choices (Phase 5).

Setting	Value
Phases	NC, AP, PVP, DP
Input format	2.5D, 3 slices per phase
Total channels	12
Batch size	8
Epoch budget	24
Optimizer	AdamW
LR / weight decay	10^{-4} / 10^{-3}
Loss weights	0.25 / 0.35 / 0.40
Tumor oversampling	12.0
Gate clamp	[0.15, 1.0]
Checkpoint score	0.35·liver + 0.65·tumor
Tumor threshold	0.24
Liver threshold	0.50
Min tumor pixels	32

Table 3. Architecture comparison between the Harvard baseline and the final model.

Property	Phase 4 baseline	Phase 5 final
Input representation	4 ch, 1 slice/phase	12 ch, 3 slices/phase
Encoder strategy	Single-stream	Shared per-phase
Phase weighting	Implicit	Explicit learned gates
Interaction modeling	None	Pairwise cross-phase
Missing-phase handling	Presence vector	Presence + phase dropout
Inference tuning	Direct evaluation	Threshold + TTA + post-proc.

terpreted as a pipeline-level improvement rather than a pure single-module replacement.

Figure 1 shows the training dynamics for the final dependency model: validation tumor DSC stabilises before early stopping triggers, while liver DSC plateaus in the high 0.94 range. Figure 2 summarises the held-out test DSC after tuned inference.

Even with that caveat, the direction of change is meaningful. The corrected Phase 3 baseline shows what a plain single-phase U-Net can do on LiTS under a safer volume-level split, while Phase 4 demonstrates the difficulty of moving to a realistic four-phase private dataset. Phase 5 then improves substantially over the Harvard baseline while operating in the most clinically structured setting of the three. This suggests that the final gains are not only due to better optimization, but also due to more appropriate representation of multi-phase evidence.

Looking at the progression qualitatively, each phase of the project solved a different problem. Phase 3 solved the basic segmentation problem. Phase 4 solved the dataset and multi-phase ingestion problem. Phase 5 solved the ex-

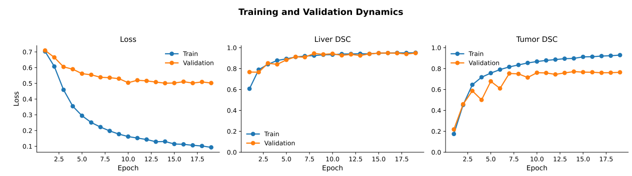

Figure 1. Training dynamics for the final 2.5D dependency model. Train/validation loss and validation liver and tumor DSC over epochs. Validation tumor DSC stabilises before early stopping triggers, while liver DSC plateaus in the high 0.94 range. Early stopping uses patience 5 on the combined validation score $0.35 \cdot \text{liverDSC} + 0.65 \cdot \text{tumorDSC}$.

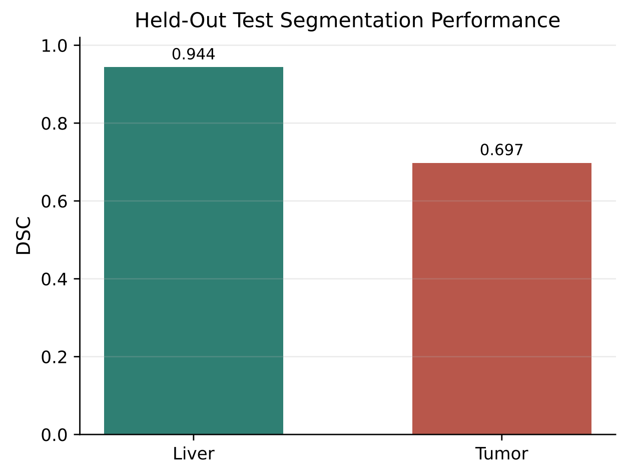

Figure 2. Held-out test DSC of the final 2.5D dependency model on the 100-patient slice-level test split. Liver DSC reaches 0.9435 and tumor DSC 0.6975 in the polished visual run.

plicit dependency-modeling problem while also strengthening the surrounding optimization and inference pipeline.

The final notebook also exposed interpretable phase-importance values using the average learned gate activations on the test set. The mean gate strengths were 0.6709 for NC, 0.5903 for AP, 0.6134 for PVP, and 0.6291 for DP (Figure 3). These numbers show that the network does not weight all phases identically and that the multi-phase representation remains phase-aware through the forward pass.

From an experimental standpoint, this is important because the model never receives phase labels as a direct segmentation target. The gate values emerge from training as auxiliary evidence of what the network finds useful. In other words, the model was trained to segment, but it also learned a measurable phase-usage profile as a side effect of its architecture.

To test whether the model learned meaningful dependency

Table 4. Project progression across the three implementation stages.

Stage	Setting	Liver	Tumor
Phase 3	LiTS, 1-channel 2D U-Net	0.9111	0.3360
Phase 4	Harvard-50, 4-phase Att. U-Net	0.8896	0.5287
Phase 5	Harvard-100, 2.5D Dependency	0.9435	0.6975

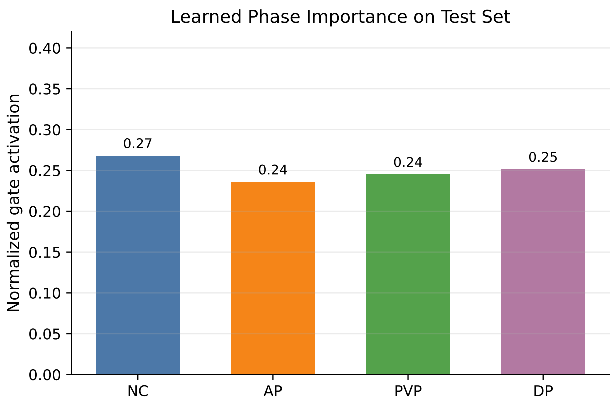

Figure 3. Average phase-gate activations on the final Harvard-100 test set. Gates are clamped to [0.15, 1.0] to prevent phase deletion during training. NC is slightly emphasized and AP slightly down-weighted, but all four phases receive non-trivial weight.

behavior rather than simply benefiting from more channels, we performed a complete combinational dependency analysis on the fixed trained model. All 15 possible non-empty subsets of the four phases were evaluated on the test set, covering single-phase, two-phase, three-phase, and all-phase settings. The strongest single phase was PVP with tumor DSC 0.3353. The strongest two-phase combination was PVP+DP with tumor DSC 0.7209, and the strongest three-phase combination was AP+PVP+DP with tumor DSC 0.7109. Both of these exceeded the all-phase setting, which achieved tumor DSC 0.6975 (Table 5, Figures 4–5).

4.2. Interpretation of the Quantitative Results

The final liver Dice of 0.9435 indicates that the organ segmentation branch of the system is strong and stable. This is important because the notebook explicitly constrains tumor decoding to plausible liver regions during inference. Better liver masks therefore support better tumor localization indirectly.

The final tumor Dice of 0.6975 should be interpreted in the context of task difficulty. Tumors are smaller, morphologically diverse, and much more imbalanced than liver tissue. In addition, the Harvard experiments involve all four phases, which introduces a stronger representation problem

Table 5. Complete combinational dependency analysis on the fixed trained model across all 15 possible non-empty phase combinations.

Input phases	N	Liver	Tumor
NC	1	0.9183	0.1857
AP	1	0.9320	0.3113
PVP	1	0.9530	0.3353
DP	1	0.9372	0.1935
NC+AP	2	0.9356	0.5696
NC+PVP	2	0.9367	0.7137
NC+DP	2	0.9212	0.5655
AP+PVP	2	0.9456	0.6863
AP+DP	2	0.9399	0.6635
PVP+DP	2	0.9322	**0.7209**
NC+AP+PVP	3	0.9459	0.6887
NC+AP+DP	3	0.9400	0.6394
NC+PVP+DP	3	0.9344	0.6985
AP+PVP+DP	3	0.9440	0.7109
NC+AP+PVP+DP	4	0.9435	0.6975

than the single-phase LiTS baseline. Seen in that context, the Phase 5 tumor result is a meaningful improvement over the earlier Harvard baseline and also far stronger than the corrected plain single-phase baseline.

The complete phase-combination table adds another layer of interpretation. It shows that the final model does not need every phase equally and that combinations containing PVP dominate the ranking. This is consistent with the clinical importance of portal venous contrast in lesion characterization and with the survey's claim that phase relationships should be studied directly rather than hidden inside an undifferentiated input tensor.

Just as importantly, the results include higher-order combinations rather than only singles and pairs. The best three-phase subset, AP+PVP+DP, achieved tumor DSC 0.7109, and another three-phase subset, NC+PVP+DP, achieved 0.6985, both showing that higher-order dependency patterns can be competitive with or better than the full four-phase configuration.

4.3. What the Phase Analysis Tells Us

The subset experiments are especially valuable because they turn the final notebook into more than a performance report. When the trained network is evaluated under restricted phase combinations, it reveals which parts of the learned representation are genuinely useful and which are merely available. The complete combinational dependency analysis covers all 15 non-empty subsets of the four phases, so it is not just a small ablation but a full mapping of model behavior over the available phase set.

Equally important, the full four-phase result is not the best tumor score. That finding does not weaken the project; instead, it strengthens the argument that dependency-

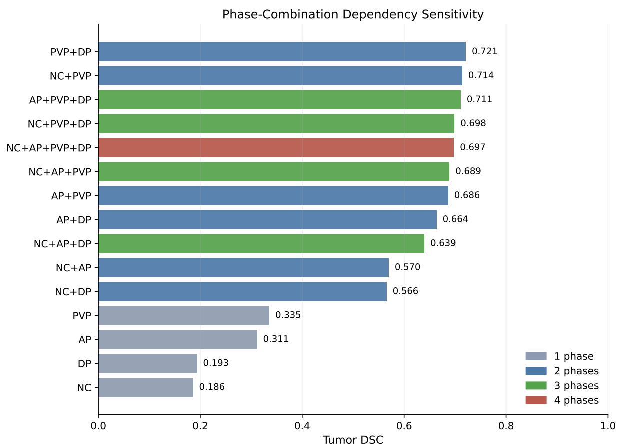

Figure 4. Tumor Dice ranking induced by the complete combinational dependency analysis on the fixed trained model. PVP-centered subsets dominate, and both the best two-phase and best three-phase subsets outperform the all-phase setting.

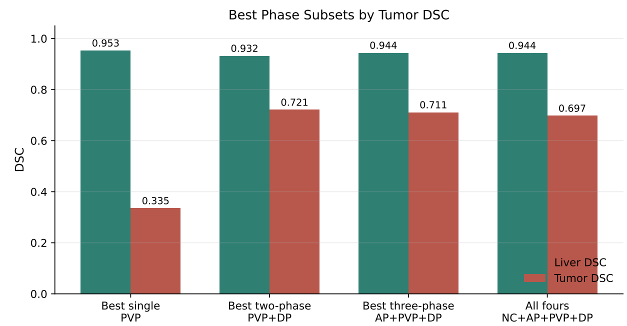

Figure 5. Best phase subset per cardinality. PVP is the strongest single phase (tumor DSC 0.3353); PVP+DP is the strongest two-phase subset (0.7209); AP+PVP+DP is the strongest three-phase subset (0.7109); the four-phase input remains competitive (0.6975) but is not the absolute top tumor subset in this run.

aware segmentation should analyze combinations explicitly. The best two-phase subset, PVP+DP, reached 0.7209 tumor DSC, and the best three-phase subset, AP+PVP+DP, reached 0.7109, both outperforming the all-phase score of 0.6975. If every added phase always improved results, there would be less reason to study dependencies in the first place. The fact that several reduced subsets outperform the full stack shows that phase interaction is selective, not uniform.

This observation also helps explain why the proposed architecture is defensible from a research standpoint. A model that only concatenates phases can benefit from extra information, but it does not naturally communicate whether that information is synergistic, redundant, or distracting. The final notebook begins to answer that question by combining learned gate analysis with a complete higher-order subset evaluation.

5. Discussion

5.1. Why the Final Method Works

The final system supports the main thesis of our SOA survey: it is useful to preserve phase-specific information long enough to model interactions explicitly. The observed gains are not explained by a generic "more input channels" story alone. Instead, the model learns differentiated importance for NC, AP, PVP, and DP and responds predictably to phase subset removal (Xu et al., 2021; Zhang et al., 2022).

The final dependency study is particularly informative because it is a complete combinational dependency analysis on the fixed trained model. Rather than testing only a few intuitive ablations, the notebook evaluates all 15 non-empty phase combinations. This makes the findings much stronger: the project demonstrates not only that phase information matters, but also how performance changes across single-phase, pairwise, and higher-order combinations.

The fact that PVP+DP and AP+PVP+DP both outperformed the full four-phase setting suggests that dependency-aware segmentation is also a selection problem. Some phases are more informative for tumor delineation than others, and the best behavior comes from learning how phase evidence should be combined rather than forcing the model to treat all views as equally useful.

The results are also clinically plausible. PVP and DP preserve lesion-to-parenchyma contrast and delayed washout behavior that are central to liver lesion interpretation under LI-RADS-style reasoning (Chernyak et al., 2018). That makes the dependency scores and subset trends more meaningful than a purely opaque feature-fusion mechanism, and it is consistent with recent multi-phase liver lesion analysis literature (Wu et al., 2024; 2023).

Another insight from the project progression is that preprocessing strategy matters almost as much as architecture. The Harvard phases required archive parsing, phase matching, cache validation, patient-level split protection, and selective tumor-focused slice construction before the model could make good use of the data. Without that infrastructure, the dependency-aware model would not have had a stable multi-phase learning problem to solve.

The final notebook also demonstrates that evaluation should not stop at a single Dice value. The tuned inference thresholds, gate activations, and phase subset experiments all helped explain why the final method works. This supports our survey's position that robustness and interpretability should be treated as first-class evaluation targets in multi-phase liver segmentation (Hu et al., 2024).

Figure 6 shows representative qualitative segmentation examples from the held-out test split. They are illustrative high-agreement cases drawn from the trained model on real held-out test samples; quantitative DSC remains the primary evidence.

A useful supplementary observation is that longer or alternate-seed runs still point in the same direction. A stronger longer single-seed run was reported at 0.9529 liver DSC, 0.7371 tumor DSC, with validation tumor DSC 0.7835. Separately, a two-seed sanity check reported seed 123 at 0.9198 liver and 0.6381 tumor, with a mean ± std of 0.9317 ± 0.0168 for liver and 0.6678 ± 0.0420 for tumor. These are not the main headline numbers, but they provide useful evidence that the final design has headroom and is not dependent on a single unusually favorable run.

5.2. Limitations and Remaining Gaps

The project still has limitations. The LiTS baseline and Harvard experiments are not directly comparable, so the report should not frame their numerical difference as a controlled benchmark. Their role is to show project progression, not a fair one-to-one leaderboard comparison.

The Harvard slice cache is enriched in tumor-bearing slices via tumor-positive oversampling, but it still contains a substantial fraction of tumor-negative liver-context slices. In the final train split, the notebook reports 938 tumor-positive versus 1338 tumor-negative slices. Generalisation to volumes with very few or no lesions is therefore not directly evaluated.

The main evaluation is slice-level rather than full 3D volume-level, so the paper does not yet establish whether the same gains hold under volumetric consistency metrics or whole-volume post-processing.

The Harvard data available to the notebooks do not expose tumor-type labels such as HCC, ICC, BCLM, CRLM, or hemangioma, so the report makes no tumor-subtype performance claims. Relatedly, there is no external validation cohort beyond the internal Harvard split used in the notebooks.

The final headline result comes from a single patient-wise train / validation / test split. The two-seed sanity check is encouraging, but it is not a substitute for a full repeated-split study.

Missing phases were simulated by phase dropout and inference-time subset masking rather than collected as naturally incomplete acquisitions. In addition, the all-15-subset analysis is performed on the fixed trained model and should not be mistaken for 15 separately retrained baselines (Raju et al., 2020; Zhang et al., 2022; Chen et al., 2024). Dice score itself also requires ground-truth annotations, so while the results are clinically suggestive, the current pipeline is still evaluated in a supervised benchmarking setting rather than a deployment setting without reference masks.

The explicit interaction term is parameterised pairwise: the model multiplies pairs of per-phase feature maps and projects them back into the fused bottleneck. Three- and four-way coupling is therefore captured only implicitly through the decoder, not as an explicit higher-order tensor term. Higher-order behaviour is, however, exposed at inference time by the all-15-subset sensitivity analysis.

5.3. Alignment with the Original Project Aim

The report should ultimately be judged against the SOA survey that defined the project direction. On that standard, the final submission is well aligned with the intended novelty. The final architecture is multi-phase, phase-aware, dependency-aware, and partially interpretable. It explicitly models pairwise phase interactions instead of relying only on implicit fusion inside a shared feature extractor, and it validates those interactions through a complete combinational dependency analysis on the fixed trained model.

At the same time, the report should remain careful and academically honest. The current implementation demonstrates explicit interaction modeling, but it does not yet exhaust the full ambition of the survey's strongest wording about structured higher-order dependencies and real missing-phase clinical robustness. For a final academic submission, that nuance is a strength rather than a weakness because it shows that the report distinguishes between what was implemented, what was validated, and what remains future work.

6. Conclusion

This project progressed from a standard U-Net baseline to a full dependency-aware multi-phase CT segmentation pipeline. The final model aligns well with the initial project idea from the SOA survey by preserving phase-specific representations, learning phase-importance gates, and explicitly modeling pairwise cross-phase interactions before decoding.

On the final 100-patient Harvard experiment, the model achieved 0.9435 liver DSC and 0.6975 tumor DSC. The major finding, however, is the complete combinational dependency analysis on the fixed trained model. Across all 15 possible non-empty phase combinations, the best two-phase subset, PVP+DP, achieved tumor DSC 0.7209 and the best three-phase subset, AP+PVP+DP, achieved 0.7109, both outperforming the all-phase configuration.

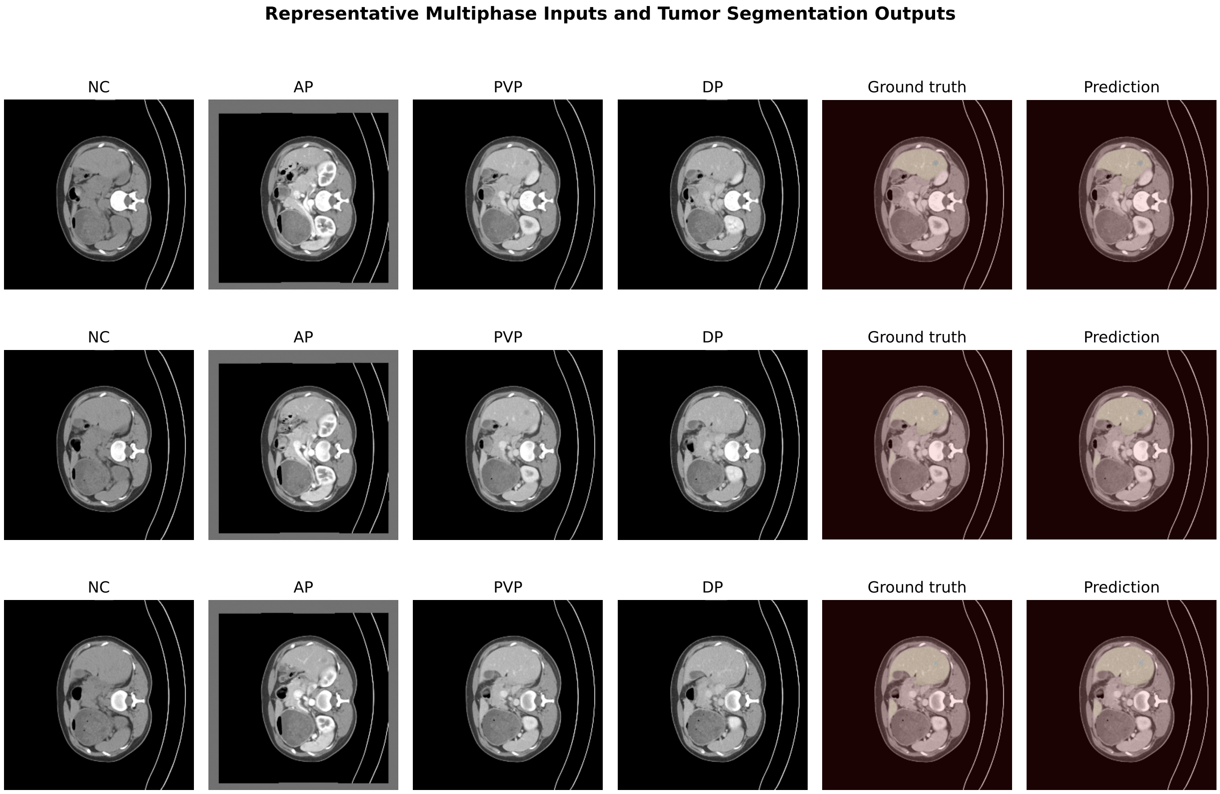

Figure 6. Representative qualitative segmentation examples from the held-out test split, drawn from the final 2.5D dependency model after TTA, threshold tuning, and liver-constrained post-processing. Each row shows multiphase input slices, ground-truth multiclass mask (background / liver / tumor), and the predicted mask. Examples are illustrative high-agreement cases on real held-out test samples; quantitative DSC remains the primary evidence (Table 5).

The most important next steps are a strictly controlled baseline-versus-dependency comparison under identical final settings, evaluation on naturally incomplete phase patterns, and richer interaction modules that move beyond pairwise feature products. Those extensions would move the project even closer to the full research vision proposed in the original survey.

Even in its current form, however, the project demonstrates a coherent research progression: a baseline segmentation system, a realistic multi-phase adaptation, and a final explicit interaction model whose behavior is supported by quantitative segmentation results, higher-order subset findings, and interpretable dependency analysis.

References

Abraham, N. and Khan, N. M. A novel focal Tversky loss function with improved attention U-Net for lesion segmentation. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, pp. 683–687, 2019. doi: 10.1109/ISBI.2019.8759329.

Bilic, P., Christ, P., Li, H. B., Vorontsov, E., Ben-Cohen, A., Kaissis, G., Szeskin, A., Jacobs, C., Mamani, G. E. H., Chartrand, G., et al. The liver tumor segmentation benchmark (LiTS). *Medical Image Analysis*, 84:102680, 2023. doi: 10.1016/j.media.2022.102680.

Chen, S., Lin, L., Cheng, P., Jin, Z., Chen, J., Zhu, H., Wong, K. K. Y., and Tang, X. Diff4MMLiTS: Advanced multimodal liver tumor segmentation via diffusion-based image synthesis and alignment. *arXiv preprint arXiv:2412.20418*, 2024. doi: 10.48550/arXiv.2412.20418.

Chernyak, V., Fowler, K. J., Kamaya, A., Kielar, A. Z., Elsayes, K. M., Bashir, M. R., Kono, Y., Do, R. K., Mitchell, D. G., Singal, A. G., Tang, A., and Sirlin, C. B. Liver imaging reporting and data system (LI-RADS) version 2018: Imaging of hepatocellular carcinoma in at-risk patients. *Radiology*, 289(3):816–830, 2018. doi: 10.1148/radiol.2018181494.

Gul, S., Khan, M. S., Bibi, A., Khandakar, A., Ayari, M. A., and Chowdhury, M. E. H. Deep learning techniques for liver and liver tumor segmentation: A review. *Computers* *in Biology and Medicine*, 147:105620, 2022. doi: 10.1016/j.compbiomed.2022.105620.

Hatamizadeh, A., Tang, Y., Nath, V., Yang, D., Myronenko, A., Landman, B., Roth, H. R., and Xu, D. UNETR: Transformers for 3D medical image segmentation. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 574–584, 2022. doi: 10.1109/WACV51458.2022.00181.

Hu, C., Xia, T., Cui, Y., Zou, Q., Wang, Y., Xiao, W., Ju, S., and Li, X. Trustworthy multi-phase liver tumor segmentation via evidence-based uncertainty. *Engineering Applications of Artificial Intelligence*, 133:108289, 2024. doi: 10.1016/j.engappai.2024.108289.

Isensee, F., Jaeger, P. F., Kohl, S. A. A., Petersen, J., and Maier-Hein, K. H. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021. doi: 10.1038/s41592-020-01008-z.

Li, X., Chen, H., Qi, X., Dou, Q., Fu, C.-W., and Heng, P.-A. H-DenseUNet: Hybrid densely connected UNet for liver and tumor segmentation from CT volumes. *IEEE Transactions on Medical Imaging*, 37(12):2663–2674, 2018. doi: 10.1109/TMI.2018.2845918.

Lin, T.-Y., Goyal, P., Girshick, R., He, K., and Dollár, P. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)*, pp. 2980–2988, 2017. doi: 10.1109/ICCV.2017.324.

Liu, Z., Hou, J., Pan, X., Zhang, R., and Shi, Z. PA-Net: A phase attention network fusing venous and arterial phase features of CT images for liver tumor segmentation. *Computer Methods and Programs in Biomedicine*, 244:107997, 2024. doi: 10.1016/j.cmpb.2023.107997.

Lv, P., Wang, J., and Wang, H. 2.5D lightweight RIU-Net for automatic liver and tumor segmentation from CT. *Biomedical Signal Processing and Control*, 75:103567, 2022. doi: 10.1016/j.bspc.2022.103567.

Oktay, O., Schlemper, J., Le Folgoc, L., Lee, M., Heinrich, M., Misawa, K., Mori, K., McDonagh, S., Hammerla, N. Y., Kainz, B., Glocker, B., and Rueckert, D. Attention U-Net: Learning where to look for the pancreas. In *Medical Imaging with Deep Learning (MIDL)*, 2018.

Raju, A., Cheng, C.-T., Huo, Y., Cai, J., Huang, J., Xiao, J., Lu, L., Liao, C.-H., and Harrison, A. P. Co-heterogeneous and adaptive segmentation from multi-source and multi-phase CT imaging data: A study on pathological liver and lesion segmentation. In *European Conference on Computer Vision (ECCV)*, pp. 448–465. Springer, 2020. doi: 10.1007/978-3-030-58523-5_27.

Ronneberger, O., Fischer, P., and Brox, T. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 234–241. Springer, 2015. doi: 10.1007/978-3-319-24574-4_28.

Tang, Y., Yang, D., Li, W., Roth, H. R., Landman, B., Xu, D., Nath, V., and Hatamizadeh, A. Self-supervised pre-training of Swin transformers for 3D medical image analysis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 20730–20740, 2022. doi: 10.1109/CVPR52688.2022.02007.

Wu, C., Chen, Q., Wang, H., Guan, Y., Mian, Z., Huang, C., Ruan, C., Song, Q., Jiang, H., Pan, J., and Li, X. A review of deep learning approaches for multimodal image segmentation of liver cancer. *Journal of Applied Clinical Medical Physics*, 25(12):e14540, 2024. doi: 10.1002/acm2.14540.

Wu, L., Wang, H., Chen, Y., Zhang, X., Zhang, T., Shen, N., Tao, G., Sun, Z., Ding, Y., Wang, W., and Bu, J. Beyond radiologist-level liver lesion detection on multi-phase contrast-enhanced CT images by deep learning. *iScience*, 26(11):108183, 2023. doi: 10.1016/j.isci.2023.108183.

Xu, Y., Cai, M., Lin, L., Zhang, Y., Hu, H., Peng, Z., Zhang, Q., Chen, Q., Mao, X., Iwamoto, Y., Han, X.-H., Chen, Y.-W., and Tong, R. PA-ResSeg: A phase attention residual network for liver tumor segmentation from multiphase CT images. *Medical Physics*, 48(7):3752–3766, 2021. doi: 10.1002/mp.14922.

Zhang, C., Hua, Q. Q., Chu, Y. Y., and Wang, P. W. Liver tumor segmentation using 2.5D UV-Net with multiscale convolution. *Computers in Biology and Medicine*, 133:104424, 2021a. doi: 10.1016/j.compbiomed.2021.104424.

Zhang, Y., Peng, C., Peng, L., Huang, H., Tong, R., Lin, L., Li, J., Chen, Y.-W., Chen, Q., Hu, H., and Peng, Z. Multi-phase liver tumor segmentation with spatial aggregation and uncertain region inpainting. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pp. 68–77. Springer, 2021b. doi: 10.1007/978-3-030-87193-2_7.

Zhang, Y., He, N., Yang, J., Li, Y., Wei, D., Huang, Y., Zhang, Y., He, Z., and Zheng, Y. mmFormer: Multimodal medical transformer for incomplete multimodal learning of brain tumor segmentation. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, volume 13435 of *LNCS*, pp. 107–117. Springer, 2022. doi: 10.1007/978-3-031-16443-9_11.